

Rodolfo Guadalupe Blanco Rodríguez

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My research focuses on modeling complex systems across multiple spatial and temporal scales using computational and mathematical approaches. I develop and apply molecular dynamics, Monte Carlo, differential equation-based, and agent-based models to problems ranging from granular materials and cell migration to epidemiological networks and pathogen transmission. In parallel, I use machine learning and data analysis techniques, including clustering and topological data analysis, to extract structure from complex datasets. My work is inherently interdisciplinary, integrating theory, computation, and collaboration across fields.

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow

Moscow, Idaho, USA
Nov 2022 - Present

University of Idaho

Department of Mathematics and Statistical Science
Supervisor: Esteban A. Hernández-Vargas

Postdoctoral Fellow

Juriquilla, Querétaro, México
Sep 2020 - Aug 2022

Universidad Nacional Autónoma de México

Institute of Mathematics
Supervisor: Esteban A. Hernández-Vargas

EDUCATION

PhD in Applied Physics

Mérida, Yucatán, México
Sep 2015 - Oct 2019

CINVESTAV

Department of Applied Physics
Dissertation: Numerical study of stress distribution in confined and flowing granular matter
Supervisor: Gabriel Guillermo Perez Angel

MSc in Applied Physics

Mérida, Yucatán, México
Sep 2013 - Aug 2015

CINVESTAV

Department of Applied Physics
Dissertation: Stress distribution in two-dimensional silos
Supervisor: Gabriel Guillermo Perez Angel

BEng in Physical Engineer

Mérida, Yucatán, México
Sep 2006 - Jun 2013

Universidad Autonoma de Yucatan

Faculty of Engineering
Dissertation: Enhancement of thermal interface materials with carbon nanofibers
Supervisor: Ruben A. Medina Esquivel

PUBLICATIONS

1. **Blanco-Rodríguez, R.**, Chi, L.A., & Hernandez-Vargas, E. A. (2026) Multicompartment Modeling of Influenza A Replication Along the Murine Respiratory Tract. *Frontiers in Virology*, 6, 1772993. <https://doi.org/10.3389/fviro.2026.1772993>
2. **Blanco-Rodríguez, R.**, Miura, T. A., & Hernandez-Vargas, E. (2025). CrossLabFit: A novel framework for integrating qualitative and quantitative data across multiple labs for model calibration. *PLOS Computational Biology*, 21(11), e1013704. <https://doi.org/10.1371/journal.pcbi.1013704>
3. Anderson, A., Kinahan, M., **Blanco-Rodríguez, R.**, Gonzalez, A. H., Udekwu, K., & Hernandez-Vargas, E. A. (2025). Computational Platform for streamlining the success of sequential antibiotic therapy. *npj Antimicrob Resist* 3, 90 (2025). <https://doi.org/10.1038/s44259-025-00160-w>
4. **Blanco-Rodríguez, R.**, Anderson, A., & Hernandez-Vargas, E. (2025). Linking within-host immune dynamics to between-host transmission and reinfection risk. *Journal of Theoretical Biology*, 112210. <https://doi.org/10.1016/j.jtbi.2025.112210>
5. **Blanco-Rodríguez, R.**, Humphrey, Alan IV, Anderson Alejandro, & Hernández-Vargas, E. (2025). Parameter Identification for PDEs via Qualitative Constraints. *IFAC-PapersOnLine*, 59(8), 72-77. <https://doi.org/10.1016/j.ifacol.2025.08.069>
6. **Blanco-Rodríguez, R.**, & Hernández-Vargas, E. (2024, September). Calibration of Stochastic Dynamic Networks: An Application to Epidemics. In *World Automation Congress* (pp. 46-57). Cham: Springer Nature Switzerland. https://doi.org/10.1007/978-3-032-10150-1_5
7. **Blanco-Rodríguez, R.**, Tetteh, J. N., & Hernández-Vargas, E. (2024). Assessing the impacts of vaccination and viral evolution in contact networks. *Scientific Reports*, 14(1), 15753. <https://doi.org/10.1038/s41598-024-66070-5>
8. Chi-Uluac, L. A., Asgharpour, S., **Blanco-Rodríguez, R. G.**, & Martínez-Archundia, M. (2023). Atomistic Molecular Insights into Angiotensin-(1-7) Interpeptide Interactions. *Journal of Chemical Information and Modeling*. <https://doi.org/10.1021/acs.jcim.3c00478>
9. **Blanco-Rodríguez, R.**, Ordoñez-Jiménez, F., Almocera, A. E. S., Chinney-Herrera, G., & Hernandez-Vargas, E. (2023). Topological data analysis of antibody dynamics of severe and non-severe patients with COVID-19. *Mathematical Biosciences*, 361, 109011. <https://doi.org/10.1016/j.mbs.2023.109011>
10. Rodríguez-Martínez, E. A., Rivas, C. U., Ayala, M. E., **Blanco-Rodríguez, R.**, Juárez, N., Hernandez-Vargas, E. A., & Aragón, A. (2023). A new computational approach, based on images

trajectories, to identify the subjacent heterogeneity of sperm to the effects of ketanserin. Cytometry Part A. <https://doi.org/10.1002/cyto.a.24732>

11. **Blanco-Rodríguez, R.**, Du, X., & Hernandez-Vargas, E. (2021). Computational simulations to dissect the cell immune response dynamics for severe and critical cases of SARS-CoV-2 infection. Computer Methods and Programs in Biomedicine, 211, 106412. <https://doi.org/10.1016/j.cmpb.2021.106412>
12. **Blanco-Rodríguez, R.**, Hidalgo, R. C., Pérez-Ángel, G., & Maza, D. (2021). Critical numerical analysis of quasi-two-dimensional silo-hopper discharging. Granular Matter, 23(4), 1-10. <https://doi.org/10.1007/s10035-021-01159-6>
13. **Blanco-Rodríguez, R.**, & Pérez-Ángel, G. (2018). Stress distribution in two-dimensional silos. Physical Review E, 97(1), 012903. <https://doi.org/10.1103/PhysRevE.97.012903>

MANUSCRIPTS IN PREPARATION

Interferon-Mediated Antiviral Protection Limits Influenza Spread: A Spatial Agent-Based Model
Expected submission: Fall 2026

A Stochastic Germinal Center Model in Antigenic Space for Cross-Variant Antibody Responses to SARS-CoV-2 Vaccination
Expected submission: Fall 2026

TEACHING EXPERIENCE

MATH 4830 - Foundations of Machine Learning

Principal Instructor

University of Idaho, Moscow, Idaho, USA

Fall 2025

MATH 4830 - Foundations of Machine Learning

Principal Instructor

University of Idaho, Moscow, Idaho, USA

Fall 2024

Thermodynamics

Co-Instructor

CFATA at the UNAM, Juriquilla, Querétaro, México

Feb 2022 - Apr 2022

Machine Learning

Teaching Assistant

Faculty of Sciences at the UNAM, Ciudad de México, México

Feb 2021 - Jun 2021

AWARDS

College of Science Travel Award

\$1000 USD awarded to support travel to SIAM Annual Meeting 2024
Jul 2024

Landahl Travel Award, Society for Mathematical Biology

\$450 USD awarded to support travel to SMB Annual Meeting 2023
Jun 2023

ACADEMIC ACTIVITIES

Thesis Committee Member

Master's Thesis Defense in Mathematics

"Modeling Mechanisms of Late-Phase and Spatial Control of Within-Host Influenza A Virus Infection" — Benjamin Whipple

Department of Mathematics and Statistical Science, University of Idaho
May 2026

Thesis Committee Member

Master's Thesis Defense in Statistics

"Multivariate Mixed Effects Model on Mouse Infection" — Brett Flerchinger

Department of Mathematics and Statistical Science, University of Idaho
April 2025

Committee Member

Undergraduate Supervised Professional Practice Defense in Biomedical Physics

"Mathematical Modeling to Decipher Multifactorial Events in Severe COVID-19 Patients" — Fernanda Ordóñez Jiménez

Faculty of Sciences, National Autonomous University of Mexico (UNAM)
July 2021

PEER REVIEW ACTIVITY

Frontier in Cellular and Infection Microbiology — 2 manuscript reviewed

Mathematical Biosciences and Engineering — 3 manuscripts reviewed

PLOS Computational Biology — 2 manuscripts reviewed

Journal of the Franklin Institute — 1 manuscript reviewed

Annual Reviews in Control — 2 manuscripts reviewed

Frontiers in Microbiology — 1 manuscript reviewed

Social Networks — 1 manuscript reviewed

PROFESSIONAL AFFILIATIONS

Active Member, Society for Mathematical Biology (SMB)

PRESENTATIONS

Conference Paper: 5th IFAC Workshop on Control of Systems Governed by PDE
June 2025, Beijing, China

Contributed Talk: SIAM Annual Meeting 2024
July 2024, Spokane, USA

Contributed Talk: SMB Annual Meeting 2023
June 2023, Columbus, USA

Contributed Talk: 2nd Annual Meeting of MexSIAM
June 2021, Online

Contributed Poster: Traffic and Granular Flow 2019
July 2019, Pamplona, Spain

Contributed Poster: XLVIII Winter Meeting on Statistical Physics 2019
January 2019, Puebla, México

Contributed Poster: 30th International Conference on Science and Technology of Complex Fluids
June 2018, San Luis Potosí, México

Contributed Poster: XLVII Winter Meeting on Statistical Physics 2018
January 2018, Puebla, México

Contributed Talk: 2nd International Workshop on Matter Out of Equilibrium
August 2016, Guanajuato, México

Contributed Poster: XLV Winter Meeting on Statistical Physics 2016
January 2016, Taxco, México

RESEARCH STAYS

Forschungszentrum Jülich

Research project: Simulation of red blood cell membrane disruption by merozoites
Jülich, Germany
January 2020 - February 2020

Universidad de Navarra

Research project: Numerical study of the effect of hopper angle on flow of granular material
Pamplona, Spain
June 2019 - August 2019